Constructing Surfaces from Cayley Graphs

Spectral Drawing, Face Systems, and Dual Graph Invariants

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1 Introduction

This thesis studies how a finite group, viewed through its Cayley graph, can give rise to a closed orientable surface. The guiding principle is that a Cayley graph provides only a one-dimensional combinatorial skeleton: it contains vertices and edges, but it does not by itself determine which cycles should be filled as faces. The missing two-dimensional datum is the face system. Thus the central object of this thesis is not merely the pair (G, S), but the triple

$$(G, S, \mathcal{F}),$$

where G is a finite group, S is a symmetric generating set, and $\mathcal{F}$ is a G-invariant collection of directed cycles satisfying a precise dart condition.

The main theme is the separation between algebraic symmetry, geometric embedding, and topological verification. The Cayley graph $\mathrm{Cay}(G, S)$ records the algebraic structure of G through right multiplication by generators. Spectral drawing uses the eigenspaces of the unweighted graph Laplacian

$$L = |S|I - A$$

to produce symmetry-preserving coordinates. However, the topology of the resulting surface is determined only after a valid face system $\mathcal{F}$ is chosen and verified. In particular, the same Cayley skeleton can support different face systems and hence produce closed orientable surfaces of different genera.

1.1 Problem Background and Mathematical Motivation

1.1.1 Cayley Skeletons and Group Combinatorics

Let G be a finite group, and let $S \subset G$ be a symmetric generating set satisfying

$$e \notin S, \qquad S = S^{-1}.$$

The Cayley graph $\mathrm{Cay}(G, S)$ is the $|S|$-regular graph whose vertex set is G and whose edges are the unordered pairs

$$\{g, gs\}, \qquad g \in G,\ s \in S.$$

Since $S = S^{-1}$, the directed steps

$$g \longrightarrow gs, \qquad gs \longrightarrow g$$

represent the same undirected edge. Thus $\mathrm{Cay}(G, S)$ is naturally an undirected regular graph.

A central point of the construction is that the Cayley graph carries two algebraically distinct operations, and these must not be confused.

Right multiplication: edge traversal. Moving along an edge of $\mathrm{Cay}(G, S)$ is governed by right multiplication. Starting from $g \in G$ and applying a generator $s \in S$ moves the vertex to gs. Therefore a word

$$w = s_1 s_2 \cdots s_k$$

is read from left to right as the walk

$$g \longmapsto gs_1 \longmapsto gs_1 s_2 \longmapsto \cdots \longmapsto gs_1 s_2 \cdots s_k.$$

If $s_1 s_2 \cdots s_k = e$, this walk closes and gives a cycle in $\mathrm{Cay}(G, S)$. Such relator cycles are the basic candidates from which faces will later be chosen.

Left multiplication: face translation. The group G also acts on the vertices of $\mathrm{Cay}(G, S)$ by left multiplication:

$$h \cdot g = hg.$$

This action preserves adjacency, because the edge $\{g, gs\}$ is sent to

$$\{hg, hgs\},$$

which is again an edge labelled by the same generator s. Hence the left action of G is an action by graph automorphisms.

The role of this left action is global rather than local. Once a cycle C based at the identity is selected as a face candidate, the whole G-orbit

$$\{h \cdot C : h \in G\}$$

must be included in the face system. This is the G-invariance requirement on $\mathcal{F}$. Thus right multiplication traces each local cycle, while left multiplication translates entire cycles across the Cayley skeleton.

This asymmetry is fundamental. Right multiplication constructs the internal sequence of vertices in a face, whereas left multiplication distributes that face consistently over the graph. The surface construction depends on keeping these two algebraic roles separate.

1.1.2 The Combinatorial Manifold Conditions and Dart Formalism

The one-dimensional graph $\mathrm{Cay}(G, S)$ becomes a closed orientable surface only after one specifies which directed cycles are to be filled as faces. This additional datum is the face system $\mathcal{F}$. To express the required closedness and orientability conditions purely combinatorially, we use darts.

Definition 1.1 (Dart)**.** A *dart* is an ordered pair

$$(g, s) \in G \times S,$$

representing the directed edge from g to gs in $\mathrm{Cay}(G, S)$. The reversal of this dart is

$$\overline{(g, s)} = (gs, s^{-1}),$$

which traverses the same undirected edge in the opposite direction.

Definition 1.2 (Face system)**.** A *face system* for $\mathrm{Cay}(G, S)$ is a finite collection $\mathcal{F}$ of directed simple cycles in $\mathrm{Cay}(G, S)$ satisfying the following conditions:

(i) every cycle in $\mathcal{F}$ is simple;

(ii) $\mathcal{F}$ is invariant under the left action of G;

(iii) every dart $(g, s) \in G \times S$ belongs to exactly one cycle in $\mathcal{F}$.

Condition (iii) is the central combinatorial constraint of the thesis. We call it the *dart condition*. It says that the face system $\mathcal{F}$ gives an exact partition of the dart set

$$\mathcal{D} = G \times S.$$

This single condition simultaneously encodes the two surface properties needed in this work.

First, each undirected edge $\{g, gs\}$ has exactly two directed realizations:

$$(g, s), \qquad (gs, s^{-1}).$$

If every dart belongs to exactly one face, then the two directed realizations of the same edge are assigned to two opposite face incidences. Thus no edge-side is left unassigned and no directed edge is used twice.

Second, the two faces adjacent along a common edge traverse that edge in opposite dart directions. This gives the combinatorial form of orientability. In particular, orientability is verified directly by the dart-reversal structure rather than by any external homological machinery.

Once $\mathcal{F}$ satisfies the dart condition, the resulting surface is a closed orientable surface with

$$V = |G|, \qquad E = \frac{|G||S|}{2}, \qquad F = |\mathcal{F}|.$$

Its Euler characteristic is

$$\chi = V - E + F.$$

By the classification of closed orientable surfaces,

$$\chi = 2 - 2g,$$

where g is the genus.

This immediately reveals the main mathematical tension. The pair (G, S) determines the Cayley skeleton and therefore fixes V and E. However, it does not determine F. The face count depends on the choice of $\mathcal{F}$. Hence the same Cayley graph may support different valid face systems and produce surfaces of different genera.

The key example is the Cayley graph of the dihedral group D_4 with

$$S = \{r, r^{-1}, s\}.$$

This graph has

$$V = 8, \qquad E = 12.$$

One valid face system has six square faces, giving

$$\chi = 8 - 12 + 6 = 2,$$

so the surface is a sphere. Another valid face system has four hexagonal faces, giving

$$\chi = 8 - 12 + 4 = 0,$$

so the surface is a torus. The Cayley skeleton is identical in both cases; only the face system changes. This example is the central motivation for separating the algebraic graph, the spectral drawing, and the combinatorial face verification.

1.2 Computational Methodology and Spectral Pipeline

1.2.1 Representation-Theoretic Graph Spectra

Let

$$\Gamma = \mathrm{Cay}(G, S)$$

and let A be its unweighted adjacency matrix. Since Γ is $|S|$-regular, the graph Laplacian used throughout this thesis is

$$L = |S|I - A.$$

No normalized Laplacian is used in this work.

The left action of G on itself induces the left regular representation on functions on G. Since left multiplication preserves adjacency, both A and L commute with this group action. Consequently, the eigenspaces of A and L are compatible with the representation theory of G.

The complex group algebra decomposes as

$$\mathbb{C}[G] \cong \bigoplus_{\rho \in \widehat{G}} V_\rho^{\oplus \deg(\rho)},$$

where $\widehat{G}$ is the set of irreducible complex representations of G, and $\deg(\rho) = \dim V_\rho$. On the representation block associated with ρ, the adjacency operator is represented by the finite matrix

$$A_\rho = \sum_{s \in S} \rho(s).$$

Therefore the spectrum of the full adjacency operator is obtained from the spectra of these smaller matrices A_ρ, and the Laplacian eigenvalues are obtained by the relation

$$\lambda_L = |S| - \lambda_A.$$

The multiplicity structure is representation-theoretic. The representation V_ρ appears $\deg(\rho)$ times inside the regular representation, so each eigenvalue coming from the matrix A_ρ appears globally with the corresponding representation multiplicity, before any accidental coincidences with eigenvalues from other representations are combined. In the special case where A_ρ acts as a scalar on V_ρ, the resulting global multiplicity is $\deg(\rho)^2$. Thus spectral degeneracy is not merely numerical; it reflects the internal representation structure of the finite group.

For spectral drawing, one chooses a real eigenspace or a direct sum of real eigenspaces and uses its basis functions as coordinate functions. If

$$W = \operatorname{span}\{\psi_1, \ldots, \psi_k\}$$

is the chosen coordinate space, then the spectral map is

$$\Phi : G \longrightarrow \mathbb{R}^k, \qquad \Phi(g) = \big(\psi_1(g), \ldots, \psi_k(g)\big).$$

Because W is invariant under the left action of G, the resulting drawing is equivariant: group symmetries appear as Euclidean symmetries of the embedded point configuration.

A three-dimensional eigenspace may suggest a spherical polyhedral realization, while a $2 + 2$ direct-sum structure of two real two-dimensional eigenspaces may suggest a toroidal realization. However, spectral drawing only suggests geometry. The actual surface type is verified only after a face system $\mathcal{F}$ satisfies the dart condition.

1.2.2 Dual Graph Spectra and Information Loss

The vertex Laplacian $L = |S|I - A$ depends only on the Cayley skeleton $\mathrm{Cay}(G, S)$. Therefore it is blind to the choice of face system $\mathcal{F}$. If the same Cayley graph admits two distinct valid face systems $\mathcal{F}_1$ and $\mathcal{F}_2$, the vertex spectrum remains unchanged:

$$\mathrm{Spec}(L_{\mathrm{Cay}(G,S)})$$

is the same for both surface systems.

To encode information depending on $\mathcal{F}$, one may pass to the classical dual graph.

Definition 1.3 (Classical dual graph)**.** Given a valid surface system $(G, S, \mathcal{F})$, the classical dual graph Γ^* is the graph whose vertices correspond to faces $F \in \mathcal{F}$. Two dual vertices are connected when the corresponding faces share an edge of the original Cayley skeleton.

The dual graph Γ^* is sensitive to the face system. For example, in the D_4 case, the spherical square-face system and the toroidal hexagon-face system have the same vertex spectrum on $\mathrm{Cay}(D_4, S)$, but they produce different dual graphs. Thus Γ^* begins to record topology that the original vertex Laplacian cannot see.

Nevertheless, the classical dual graph loses two important types of information. First, it forgets the original Cayley vertices, since each face is collapsed into a single dual vertex. Second, it forgets the cyclic dart order inside each face. It records that two faces are adjacent, but not how the vertices and darts are cyclically arranged around each face.

This motivates the face-inflated graph, denoted

$$\Gamma_{\mathrm{inf}}.$$

Instead of collapsing each face to a single vertex, Γ_{inf} replaces each face by a small ordered polygon. The vertices of this polygon are face-local copies of the original Cayley vertices, and the polygon edges record the cyclic dart order of the face. Radial incidence edges then connect each face-local copy back to the corresponding original Cayley vertex.

Thus Γ_{inf} retains three pieces of information simultaneously:

$$\mathrm{Cay}(G, S), \qquad \mathcal{F}, \qquad \text{the cyclic dart order of each face.}$$

This makes it a more faithful graph-theoretic data structure for studying a chosen surface system spectrally.

1.2.3 Summary of the Eight-Step Operational Triad

The methodology of the thesis is organized around the principle:

spectral drawing suggests; face systems verify; dual-inflated graphs record.

This principle can be implemented as the following eight-step pipeline.

(1) **Choose algebraic data.** Select a finite group G and a symmetric generating set $S = S^{-1}$ with $e \notin S$.

(2) **Construct the Cayley skeleton.** Build the graph $\mathrm{Cay}(G, S)$ and record

$$V = |G|, \qquad E = \frac{|G||S|}{2}.$$

(3) **Form the unweighted Laplacian.** Construct the adjacency matrix A and the graph Laplacian

$$L = |S|I - A.$$

(4) **Compute spectral coordinates.** Use low-frequency eigenspaces of L to define a spectral drawing

$$\Phi : G \to \mathbb{R}^k.$$

The embedding space is built from direct sums of invariant eigenspaces.

(5) **Identify candidate face cycles.** Use relator words and the geometry of the spectral drawing to select simple directed cycles as possible faces.

(6) **Generate the G-invariant face system.** Translate each chosen base cycle by the left action of G and form a candidate face system $\mathcal{F}$.

(7) **Verify the dart condition.** Check that every dart $(g, s) \in G \times S$ occurs in exactly one face of $\mathcal{F}$. This verifies that the assembled object is a closed orientable surface.

(8) **Record the chosen surface system.** Compute

$$\chi = V - E + F$$

and determine the genus from $\chi = 2 - 2g$. Then construct Γ^* or, more faithfully, Γ_{inf}, to encode the face system for further spectral analysis.

In this framework, the roles of the three main objects are sharply separated. The Cayley graph $\text{Cay}(G, S)$ supplies the symmetric one-dimensional skeleton. The face system $\mathcal{F}$ verifies the topology through the dart condition and the Euler characteristic. The face-inflated graph Γ_{inf} records the chosen surface system without discarding vertex identity or cyclic dart order.

This separation is the conceptual foundation of the thesis. The graph alone does not determine the surface. Spectral drawing alone does not verify topology. The face system is the decisive combinatorial datum, and the inflated graph is the refined structure that allows this datum to be studied within the same spectral framework.

2 Algebraic Foundations: Finite Group Actions and Combinatorial Skeletons

This section establishes the algebraic language used throughout the thesis. The goal is to separate three levels of structure: relator words inside the finite group G, the Cayley skeleton $\text{Cay}(G, S)$ determined by right multiplication, and the left G-action that translates cycles globally across the graph. This separation is essential for the later construction of G-invariant face systems.

2.1 Group Presentations and Relator Words

Throughout this thesis, G denotes a finite group. We fix a symmetric generating set

$$S = S^{-1}, \qquad e \notin S,$$

and we regard words in the alphabet S as instructions for moving along the Cayley graph by right multiplication.

A group presentation is written as

$$G = \langle S \mid R \rangle,$$

where R is a collection of relator words. A relator word is a finite sequence of generators whose product is the identity element of G.

Definition 2.1 (Word and relator)**.** A *word* of length k over S is an expression

$$w = s_1 s_2 \cdots s_k, \qquad s_i \in S.$$

The word evaluates in G to the product

$$s_1 s_2 \cdots s_k \in G.$$

A word

$$r = s_1 s_2 \cdots s_k$$

is called a *relator* if

$$s_1 s_2 \cdots s_k = e$$

in G.

The convention is that words are read from left to right. Starting at a group element $g \in G$, the word

$$w = s_1 s_2 \cdots s_k$$

traces the sequence

$$g, \quad gs_1, \quad gs_1 s_2, \quad \ldots, \quad gs_1 s_2 \cdots s_k.$$

If w is a relator, then the final element is again g, so the word traces a closed walk in the Cayley graph.

Definition 2.2 (Relator cycle)**.** Let

$$r = s_1 s_2 \cdots s_k$$

be a relator. The corresponding relator cycle based at $g \in G$ is the cyclic sequence

$$C_g = \big(g,\, gs_1,\, gs_1 s_2,\, \ldots,\, gs_1 s_2 \cdots s_{k-1}\big).$$

The final step is understood to return from $gs_1 s_2 \cdots s_{k-1}$ to g by multiplying on the right by s_k.

Not every relator is suitable for defining a face. A relator may repeat vertices before returning to its starting point, in which case it gives a closed walk but not a simple cycle.

Remark 2.1 (Simple relator cycles)**.** A relator cycle

$$C_g = \big(g,\, gs_1,\, gs_1 s_2,\, \ldots,\, gs_1 s_2 \cdots s_{k-1}\big)$$

is called *simple* if all displayed vertices are pairwise distinct. Only simple relator cycles are used as face candidates in this thesis. This condition ensures that each candidate face has a well-defined cyclic dart order and does not repeat vertices internally.

Thus relator words provide the algebraic source of candidate faces, but they do not by themselves determine a surface. A surface arises only after suitable relator cycles are selected, translated by the left G-action, and verified by the dart condition.

2.2 Cayley Skeletons and Graph Operators

We now pass from words in the group to the graph-theoretic skeleton on which the surface construction will take place.

Definition 2.3 (Cayley graph)**.** Let G be a finite group and let $S \subset G$ be a symmetric generating set satisfying

$$S = S^{-1}, \qquad e \notin S.$$

The *Cayley graph* $\mathrm{Cay}(G, S)$ is the graph

$$\Gamma = (V, E)$$

defined by

$$V = \{v_g : g \in G\},$$

and

$$E = \big\{\{v_g, v_{gs}\} : g \in G,\ s \in S\big\}.$$

Each edge is produced by right multiplication by a generator s.

Since $S = S^{-1}$, the directed step

$$v_g \longrightarrow v_{gs}$$

and the directed step

$$v_{gs} \longrightarrow v_g$$

represent the same undirected edge. Hence $\mathrm{Cay}(G, S)$ is an undirected regular graph. Its number of vertices and edges is

$$|V| = |G|, \qquad |E| = \frac{|G||S|}{2}.$$

The graph is $|S|$-regular because from every vertex v_g there is one edge corresponding to each generator $s \in S$.

Definition 2.4 (Graph operators)**.** Let

$$\Gamma = \mathrm{Cay}(G, S)$$

and let $d = |S|$. The *adjacency matrix* of Γ is the unweighted matrix A indexed by $G \times G$ and defined by

$$A_{g,h} = \begin{cases} 1, & \text{if } h = gs \text{ for some } s \in S, \\ 0, & \text{otherwise.} \end{cases}$$

Since Γ is d-regular, its degree matrix is

$$D = dI = |S|I.$$

The graph Laplacian used throughout this thesis is the unweighted Laplacian

$$L = D - A = |S|I - A.$$

No normalized Laplacian is used in this work. The operator

$$L = |S|I - A$$

is the fixed vertex Laplacian associated only with the Cayley skeleton $\mathrm{Cay}(G, S)$. In particular, L depends on G and S, but not on any choice of face system $\mathcal{F}$.

Remark 2.2 (Left vertex-transitivity)**.** The group G acts on the vertex set of $\mathrm{Cay}(G, S)$ by left multiplication:

$$h \cdot v_g = v_{hg}.$$

This action preserves adjacency, since

$$\{v_g, v_{gs}\} \longmapsto \{v_{hg}, v_{hgs}\},$$

which is again an edge labelled by the same generator s. Thus every $h \in G$ acts as a graph automorphism of $\mathrm{Cay}(G, S)$. The action is free and transitive on vertices, so $\mathrm{Cay}(G, S)$ is vertex-transitive.

This vertex-transitivity is the graph-theoretic expression of the homogeneity of G. It is also the reason why many local checks, such as the dart condition, can be verified at the identity and then propagated to every vertex by left multiplication.

The adjacency matrix A and the Laplacian $L = |S|I - A$ commute with the left action of G. Therefore their eigenspaces are compatible with the representation-theoretic decomposition of functions on G. This observation is the algebraic foundation for the spectral drawing construction developed in Section 4.

2.3 The Dart Algebra and Disjoint Algebraic Actions

The Cayley graph $\mathrm{Cay}(G, S)$ is undirected, but the construction of faces requires oriented edge data. For this reason, we work with darts.

Definition 2.5 (Dart)**.** A *dart* of $\mathrm{Cay}(G, S)$ is an ordered pair

$$(g, s) \in G \times S.$$

It represents the directed edge

$$v_g \longrightarrow v_{gs}.$$

The set of all darts is

$$\mathcal{D} = G \times S.$$

Hence

$$|\mathcal{D}| = |G||S| = 2|E|.$$

Each undirected edge corresponds to exactly two darts. These two darts are related by a reversal operation.

Definition 2.6 (Dart reversal)**.** The *reversal* of a dart (g, s) is

$$\overline{(g, s)} = (gs, s^{-1}).$$

This is the directed edge

$$v_{gs} \longrightarrow v_g.$$

The reversal map is an involution:

$$\overline{\overline{(g, s)}} = (g, s).$$

The dart set $\mathcal{D} = G \times S$ carries two algebraically distinct operations. The first is right multiplication, which traces edges and cycles. The second is left multiplication, which translates whole cycles. The surface construction depends on keeping these two operations separate.

Remark 2.3 (Right multiplication versus left multiplication)**.** There are two different roles played by multiplication in this thesis.

Right multiplication: local edge traversal. A dart

$$(g, s)$$

means that one moves from g to gs by multiplying on the right by s. Therefore a sequence of generators

$$s_1, s_2, \ldots, s_k$$

traces the walk

$$g \longrightarrow gs_1 \longrightarrow gs_1s_2 \longrightarrow \cdots \longrightarrow gs_1s_2 \cdots s_k.$$

If $s_1s_2 \cdots s_k = e$, this walk returns to g and gives a relator cycle.

Left multiplication: global face translation. The group G acts on darts by

$$h \cdot (g, s) = (hg, s).$$

This operation changes the base vertex but preserves the generator label. If

$$C_e = (e, s_1, s_1s_2, \ldots, s_1s_2 \cdots s_{k-1})$$

is a relator cycle based at the identity, then left multiplication by h gives

$$h \cdot C_e = (h, hs_1, hs_1s_2, \ldots, hs_1s_2 \cdots s_{k-1}).$$

Thus the left action moves the entire cycle rigidly to a new basepoint.

Separation of roles. Right multiplication constructs the internal cyclic order of a face. Left multiplication distributes that face across the Cayley skeleton. The former is local and stepwise; the latter is global and equivariant. These two operations are not interchangeable.

The compatibility between left multiplication and dart reversal is important. For any $h \in G$ and any dart (g, s), one has

$$h \cdot \overline{(g, s)} = h \cdot (gs, s^{-1}) = (hgs, s^{-1}),$$

while

$$\overline{h \cdot (g, s)} = \overline{(hg, s)} = (hgs, s^{-1}).$$

Hence

$$h \cdot \overline{(g, s)} = \overline{h \cdot (g, s)}.$$

That is, left multiplication commutes with dart reversal.

This compatibility ensures that if a face system is invariant under the left action of G, then its dart-reversal structure is transported uniformly across the graph. Consequently, the local dart condition can be checked at the identity and then extended to all vertices by left-transitivity.

The algebraic mechanism can be summarized as follows. A relator word

$$r = s_1 s_2 \cdots s_k = e$$

defines a cycle C_e based at the identity by right multiplication. The left orbit

$$\mathcal{O}(C_e) = \{h \cdot C_e : h \in G\}$$

then gives a G-invariant family of candidate faces. If

$$\mathrm{Stab}_G(C_e) = \{h \in G : h \cdot C_e = C_e\}$$

is the stabilizer of C_e, then the orbit-stabilizer theorem gives

$$|\mathcal{O}(C_e)| = \frac{|G|}{|\,\mathrm{Stab}_G(C_e)|}.$$

Thus the construction of a face system is algebraic but not automatic. One must choose relator cycles, take their left G-orbits, and then verify that the resulting collection of directed cycles gives an exact partition of the dart set $\mathcal{D}$. This dart partition is the basic combinatorial object from which the closed orientable surface will be obtained in Section 3.

3 Combinatorial Topology of Face Systems and Dart Formalism

This section gives the topological verification mechanism of the thesis. The Cayley graph $\mathrm{Cay}(G, S)$ supplies only a one-dimensional skeleton. A closed orientable surface is obtained only after a valid face system $\mathcal{F}$ is specified. The central point is that this verification can be carried out entirely through darts: every directed edge must appear in exactly one face cycle. This exact dart partition is the combinatorial condition that replaces any need for boundary-operator or chain-complex language.

Throughout this section, we fix

$$\Gamma = \mathrm{Cay}(G, S), \qquad \mathcal{D} = G \times S,$$

where $\mathcal{D}$ is the dart set of Γ.

3.1 Regular Face Systems and Orbit Decomposition

We first define the class of face systems used in this thesis. The definition has two independent components: each individual face must be a simple directed cycle, and the collection of faces must be invariant under the left action of G.

Definition 3.1 (Simple cycle)**.** A *simple cycle* in $\Gamma = \mathrm{Cay}(G, S)$ of length k based at $g \in G$ is a cyclic sequence

$$C_g = \big(g,\, gs_1,\, gs_1s_2,\, \ldots,\, gs_1s_2\cdots s_{k-1}\big),$$

where $s_1, \ldots, s_k \in S$ and

$$s_1s_2\cdots s_k = e.$$

The displayed vertices are required to be pairwise distinct. The final step returns from $gs_1s_2\cdots s_{k-1}$ to g by right multiplication by s_k.

The corresponding cyclic sequence of darts is

$$\big((g, s_1),\, (gs_1, s_2),\, \ldots,\, (gs_1s_2\cdots s_{k-1}, s_k)\big).$$

This dart sequence records both the vertices of the cycle and the clockwise or counterclockwise order in which the cycle is traversed.

Definition 3.2 (Face system)**.** A *face system* for $\Gamma = \mathrm{Cay}(G, S)$ is a finite collection $\mathcal{F}$ of simple directed cycles in Γ, whose elements are called *faces*, satisfying the following conditions:

(i) **Simplicity.** Every face in $\mathcal{F}$ is a simple cycle.

(ii) **Left G-invariance.** If

$$C_e = (e, s_1, s_1s_2, \ldots, s_1s_2\cdots s_{k-1})$$

is a face in $\mathcal{F}$, then for every $h \in G$ the left translate

$$h \cdot C_e = (h, hs_1, hs_1s_2, \ldots, hs_1s_2\cdots s_{k-1})$$

is also a face in $\mathcal{F}$.

The left G-invariance condition is essential. It ensures that the left action of G on the Cayley skeleton extends from vertices and edges to the chosen faces. Thus the group symmetry is not merely a symmetry of the graph; it is also a symmetry of the assembled surface system.

Orbit decomposition of face systems. Let C_e be a simple cycle based at the identity. Its face orbit under the left action of G is

$$\mathcal{O}(C_e) = \{h \cdot C_e : h \in G\}.$$

The stabilizer of C_e is

$$\mathrm{Stab}_G(C_e) = \{h \in G : h \cdot C_e = C_e\}.$$

By the orbit-stabilizer theorem,

$$|\mathcal{O}(C_e)| = \frac{|G|}{|\operatorname{Stab}_G(C_e)|}.$$

A G-invariant face system is therefore a disjoint union of finitely many face orbits:

$$\mathcal{F} = \mathcal{O}(C_e^{(1)}) \sqcup \mathcal{O}(C_e^{(2)}) \sqcup \cdots \sqcup \mathcal{O}(C_e^{(m)}).$$

Consequently, its total number of faces is

$$|\mathcal{F}| = \sum_{i=1}^{m} \frac{|G|}{|\operatorname{Stab}_G(C_e^{(i)})|}.$$

This orbit decomposition reduces the construction of $\mathcal{F}$ to a finite list of identity-based cycles. Once these base cycles are chosen, the left action of G determines all their translates. The remaining task is to verify that the resulting directed cycles fit together correctly. This is exactly the role of the dart condition.

3.2 The Dart Adjacency Condition for Orientable 2-Manifolds

Each face in $\mathcal{F}$ contributes a cyclic list of darts. The decisive question is whether these cyclic lists collectively use the dart set

$$\mathcal{D} = G \times S$$

in the correct way.

Definition 3.3 (Dart partition)**.** A face system $\mathcal{F}$ *partitions the dart set* $\mathcal{D}$ if every dart $(g, s) \in \mathcal{D}$ occurs in the dart sequence of at least one face in $\mathcal{F}$. It *exactly partitions* $\mathcal{D}$ if every dart occurs in the dart sequence of exactly one face in $\mathcal{F}$.

Theorem 3.1 (Dart Adjacency Condition)**.** Let $\mathcal{F}$ be a G-invariant face system for $\Gamma = \operatorname{Cay}(G, S)$ consisting of simple cycles. Then $\mathcal{F}$ defines a closed orientable surface with one-dimensional skeleton Γ if and only if $\mathcal{F}$ exactly partitions the dart set

$$\mathcal{D} = G \times S.$$

Equivalently, every dart $(g, s) \in G \times S$ must belong to exactly one face of $\mathcal{F}$.

Proof. We prove both directions.

Necessity. Assume that $\mathcal{F}$ defines a closed orientable surface with skeleton Γ. Consider any undirected edge

$$\{g, gs\}.$$

It has precisely two dart realizations:

$$(g, s), \qquad (gs, s^{-1}).$$

Since the surface is closed and orientable, each edge is adjacent to two face incidences, and these two incidences traverse the edge in opposite directions. Hence one face uses the dart

(g, s) and the other uses the reversed dart (gs, s^{-1}). Therefore every dart occurs in exactly one face. Thus $\mathcal{F}$ exactly partitions $\mathcal{D}$.

Sufficiency. Conversely, assume that $\mathcal{F}$ exactly partitions $\mathcal{D}$. Fix an undirected edge

$$\{g, gs\}.$$

Its two dart realizations are

$$(g, s) \quad \text{and} \quad (gs, s^{-1}).$$

By the exact partition hypothesis, the dart (g, s) occurs in a unique face $F_1 \in \mathcal{F}$, and the reversed dart (gs, s^{-1}) occurs in a unique face $F_2 \in \mathcal{F}$.

These two face incidences traverse the same edge in opposite directions. Thus every edge is paired by two opposite dart usages. No directed edge-side is missing, and no directed edge-side is used more than once. Therefore the local edge structure is that of a closed surface.

Moreover, because the two usages of each edge are opposite in dart direction, the orientations induced by adjacent faces are compatible across every shared edge. Assigning each face the orientation determined by its dart sequence therefore gives a consistent global orientation. Hence the surface assembled from $\mathcal{F}$ is orientable.

Finally, since $\mathcal{F}$ is invariant under the left action of G, this surface structure is transported equivariantly across the entire Cayley skeleton. Thus $\mathcal{F}$ defines a closed orientable surface with one-dimensional skeleton Γ. □

Remark 3.1 (Local verification at the identity)**.** Because the left action of G is transitive on vertices and preserves dart reversal,

$$h \cdot \overline{(g, s)} = \overline{h \cdot (g, s)},$$

it is often enough in computations to verify the dart condition at the identity vertex. Once the darts based at e are covered exactly once, left G-invariance propagates the same condition to every vertex.

Remark 3.2 (Failure modes)**.** There are two basic ways in which a proposed face system can fail the dart condition.

(i) If a dart occurs in no face, then one directed edge-side has not been assigned to any face. The proposed surface system is incomplete.

(ii) If a dart occurs in two or more faces, then the same directed edge-side is used repeatedly. The proposed surface system is overfilled along that dart direction.

Both failures are detected purely by counting dart occurrences. This is why the dart formalism is the correct combinatorial language for the surface verification used in this thesis.

3.3 Topological Invariants: Euler Characteristic and Genus Classification

Once the dart condition has been verified, the topological type of the resulting closed orientable surface is determined by the Euler characteristic.

Definition 3.4 (Euler characteristic)**.** Let $\mathcal{F}$ be a G-invariant face system satisfying the Dart Adjacency Condition. The Euler characteristic of the resulting surface is

$$\chi = V - E + F,$$

where

$$V = |G|, \qquad E = \frac{|G||S|}{2}, \qquad F = |\mathcal{F}|.$$

Using the orbit decomposition of $\mathcal{F}$, the face count can be written as

$$F = \sum_{i=1}^{m} \frac{|G|}{|\operatorname{Stab}_G(C_e^{(i)})|}.$$

The quantities V and E are fixed once the pair (G, S) is fixed. The quantity F, however, depends on the choice of the face system $\mathcal{F}$. Therefore the Euler characteristic is not an invariant of the Cayley graph alone; it is an invariant of the triple

$$(G, S, \mathcal{F}).$$

Theorem 3.2 (Genus classification)**.** Let Σ be the closed orientable surface obtained from a valid triple $(G, S, \mathcal{F})$. Then there exists a unique integer $g \geq 0$ such that

$$\chi(\Sigma) = 2 - 2g.$$

Equivalently,

$$g = 1 - \frac{\chi(\Sigma)}{2} = 1 - \frac{V - E + F}{2}.$$

The cases are:

$$g = 0 \iff \Sigma \cong S^2, \qquad g = 1 \iff \Sigma \cong T^2,$$

and $g \geq 2$ corresponds to a closed orientable surface of higher genus.

Proof. By Theorem 3.1, the face system $\mathcal{F}$ assembles the Cayley skeleton into a closed orientable surface. The classification of closed orientable surfaces states that every such surface is classified up to homeomorphism by its genus g, and its Euler characteristic is

$$\chi = 2 - 2g.$$

Solving for g gives

$$g = 1 - \frac{\chi}{2}.$$

Substituting $\chi = V - E + F$ gives the stated formula. □

The following example is the basic topological switching phenomenon driving the thesis.

Example 3.1 (D_4: one skeleton, two surface systems)**.** Let

$$G = D_4 = \langle r, s \mid r^4 = e,\ s^2 = e,\ srs = r^{-1} \rangle$$

and choose the symmetric generating set

$$S = \{r, r^{-1}, s\}.$$

Then

$$|G| = 8, \qquad |S| = 3,$$

so

$$V = 8, \qquad E = \frac{8 \cdot 3}{2} = 12.$$

Square-face system. One valid G-invariant face system consists of six square faces. Hence

$$F = 6, \qquad \chi = 8 - 12 + 6 = 2.$$

By Theorem 3.2, the genus is

$$g = 1 - \frac{2}{2} = 0.$$

Thus the resulting surface is the sphere S^2.

Hexagon-face system. Another valid G-invariant face system on the same Cayley skeleton consists of four hexagonal faces. Hence

$$F = 4, \qquad \chi = 8 - 12 + 4 = 0.$$

Therefore

$$g = 1 - \frac{0}{2} = 1.$$

The resulting surface is the torus T^2.

In both cases, the Cayley graph $\mathrm{Cay}(D_4, S)$ is identical. The vertex set, edge set, adjacency matrix, and unweighted Laplacian

$$L = |S|I - A$$

are unchanged. Only the face system $\mathcal{F}$ changes. Consequently, the change in genus is a purely combinatorial effect of choosing different valid dart partitions of the same dart set.

This example shows why the topology cannot be recovered from the Cayley skeleton alone. The graph $\mathrm{Cay}(G, S)$ determines V and E, but the choice of $\mathcal{F}$ determines F, and hence determines the genus. The subsequent sections use spectral drawing to suggest natural geometric realizations, but the final topological verification always returns to the dart condition and the Euler characteristic of the triple

$$(G, S, \mathcal{F}).$$

4 Spectral Drawing: Group Representations, Graph Spectra, and Geometric Embeddings

The previous sections established the combinatorial mechanism by which a Cayley skeleton $\mathrm{Cay}(G, S)$ becomes a closed orientable surface after a valid face system $\mathcal{F}$ is chosen. We now turn to the geometric side of the thesis: how can the abstract vertices of $\mathrm{Cay}(G, S)$ be placed in Euclidean space in a way that reflects the symmetry of G?

The method used in this thesis is *spectral drawing*. Starting from the unweighted graph Laplacian

$$L = |S|I - A,$$

we use selected eigenspaces of L as coordinate spaces. Since the Cayley graph is invariant under the left action of G, these eigenspaces are compatible with the representation theory of G. Thus spectral drawing provides a bridge from finite group symmetry to discrete geometric realization.

The central warning remains the same as before: spectral drawing suggests geometry, but it does not determine topology. The actual surface type is verified only after a face system $\mathcal{F}$ satisfies the dart condition and the Euler characteristic

$$\chi = V - E + F$$

is computed.

4.1 Representation-Theoretic Diagonalization of the Graph Laplacian

Let

$$\Gamma = \mathrm{Cay}(G, S)$$

be the Cayley skeleton of a finite group G with symmetric generating set $S = S^{-1}$ and $e \notin S$. Let A denote the unweighted adjacency matrix of Γ. Since Γ is $|S|$-regular, the Laplacian used throughout this thesis is

$$L = |S|I - A.$$

This is the ordinary unweighted graph Laplacian; no normalized Laplacian is used.

The left action of G on itself induces an action on functions $f : G \to \mathbb{C}$ by

$$(h \cdot f)(g) = f(h^{-1}g).$$

This is the left regular representation on $\mathbb{C}[G]$. Because left multiplication preserves adjacency in $\mathrm{Cay}(G, S)$, the adjacency operator A commutes with this left G-action. Consequently, the Laplacian $L = |S|I - A$ also commutes with the same action.

The representation-theoretic consequence is that the spectral structure of A and L can be analyzed through the irreducible representations of G. The group algebra decomposes as

$$\mathbb{C}[G] \cong \bigoplus_{\rho \in \widehat{G}} V_\rho^{\oplus \deg(\rho)},$$

where $\widehat{G}$ denotes the set of irreducible complex representations of G, and $\deg(\rho) = \dim V_\rho$.

For each irreducible representation ρ, define the representation block

$$A_\rho = \sum_{s \in S} \rho(s).$$

The eigenvalues of the full adjacency operator A are obtained from the eigenvalues of these smaller matrices A_ρ. If λ_A is an eigenvalue arising from the block A_ρ, then the corresponding Laplacian eigenvalue is

$$\lambda_L = |S| - \lambda_A.$$

The multiplicity is governed by the regular representation. Since V_ρ appears $\deg(\rho)$ times in $\mathbb{C}[G]$, each eigenvalue arising from A_ρ contributes to the full spectrum with this representation-theoretic repetition, before possible coincidences with eigenvalues from other irreducible representations are combined. In the special case where A_ρ acts as a scalar on V_ρ, this produces a global multiplicity of $\deg(\rho)^2$. In general, however, one should not assume that A_ρ is scalar; the correct object is the finite matrix

$$A_\rho = \sum_{s \in S} \rho(s).$$

Thus the spectral data of $\mathrm{Cay}(G, S)$ are rigidly determined by the pair (G, S). They do not depend on the face system $\mathcal{F}$. This is the source of both the power and the limitation of spectral drawing: it captures the symmetry of the Cayley skeleton, but it remains blind to the later topological choice of faces.

4.2 The Spectral Embedding Mechanism via Direct Sums of Eigenspaces

A spectral drawing assigns Euclidean coordinates to the vertices of $\mathrm{Cay}(G, S)$ by using eigenvectors of the unweighted Laplacian

$$L = |S|I - A.$$

Let

$$W \subseteq \ell^2(G)$$

be a real eigenspace, or a direct sum of real eigenspaces, chosen as the coordinate space for the drawing. Suppose

$$W = \mathrm{span}\{\psi_1, \psi_2, \ldots, \psi_k\}.$$

Then the spectral embedding is the map

$$\Phi : G \longrightarrow \mathbb{R}^k$$

defined by

$$\Phi(g) = \big(\psi_1(g), \psi_2(g), \ldots, \psi_k(g)\big).$$

Definition 4.1 (Spectral drawing)**.** Let $W \subseteq \ell^2(G)$ be a k-dimensional real coordinate space obtained from eigenspaces of the unweighted Laplacian $L = |S|I - A$. If

$$W = \operatorname{span}\{\psi_1, \ldots, \psi_k\},$$

then the associated *spectral drawing* is

$$\Phi : G \to \mathbb{R}^k, \qquad \Phi(g) = \big(\psi_1(g), \ldots, \psi_k(g)\big).$$

Because the eigenspaces of L are compatible with the left action of G, the coordinate space W may be chosen to be invariant under this action. Then the spectral drawing is equivariant in the following sense: for each $h \in G$, there exists an orthogonal transformation O_h of $\mathbb{R}^k$ such that

$$\Phi(hg) = O_h \Phi(g).$$

Thus the algebraic symmetry of G appears as Euclidean symmetry of the embedded point configuration.

Remark 4.1 (Direct-sum structure of the coordinate space)**.** The coordinate space used for a three-dimensional spectral drawing need not be an irreducible three-dimensional representation. In many examples, it is a reducible invariant space built as a direct sum of lower-dimensional representation components:

$$W = V_{\rho_1} \oplus V_{\rho_2} \oplus \cdots \oplus V_{\rho_m}.$$

The symbol $\oplus$ is essential: the coordinates are assembled by placing independent invariant components side by side as coordinate directions. This direct-sum construction is the algebraic source of the embedding space used in the examples below.

This distinction is geometrically important. A three-dimensional coordinate space may suggest a spherical polyhedral realization, while a direct sum of two real two-dimensional eigenspaces may suggest two independent circular phase directions and hence a toroidal realization. Nevertheless, these are geometric suggestions only. The topology is not settled until a valid face system $\mathcal{F}$ is chosen and verified by the dart condition.

4.3 Concrete Case Studies: Abelian Torus and Non-Abelian Combinatorial Verifications

We now examine two concrete examples. The first, $\mathbb{Z}_3 \times \mathbb{Z}_3$, shows how Fourier eigenspaces naturally produce a toroidal parameter system. The second, D_4, shows that even when the spectral embedding is fixed, different valid face systems on the same Cayley skeleton can produce different topological surfaces.

4.3.1 The Smooth Fourier Parameterization of $\mathbb{Z}_3 \times \mathbb{Z}_3$

Let

$$G = \mathbb{Z}_3 \times \mathbb{Z}_3, \qquad S = \{(\pm 1, 0), (0, \pm 1)\}.$$

Then $\mathrm{Cay}(G, S)$ is the 3×3 periodic square grid. The graph has

$$V = |G| = 9, \qquad E = \frac{|G||S|}{2} = \frac{9 \cdot 4}{2} = 18.$$

The unweighted Laplacian is

$$L = |S|I - A = 4I - A.$$

Since G is abelian, all irreducible representations are one-dimensional characters. The low-frequency eigenvectors are therefore discrete Fourier modes. The two cyclic directions of $\mathbb{Z}_3 \times \mathbb{Z}_3$ give two independent real two-dimensional coordinate systems. Algebraically, the relevant coordinate structure is

$$\mathbb{R}^2 \oplus \mathbb{R}^2.$$

The first copy of $\mathbb{R}^2$ records the Fourier phase along the first cyclic direction, and the second copy records the Fourier phase along the second cyclic direction.

Let these two phase variables be denoted by

$$\theta, \qquad \phi.$$

They should be interpreted as the two circular coordinates of the periodic grid. Thus the spectral data naturally suggest the product of two circles:

$$S^1 \times S^1,$$

which is the torus.

After subdividing and smoothing the square cells of the periodic grid, the corresponding smooth toroidal surface in $\mathbb{R}^3$ can be represented by the standard parameterization

$$X = (R + r\cos\phi)\cos\theta,$$

$$Y = (R + r\cos\phi)\sin\theta,$$

$$Z = r\sin\phi.$$

This smooth model is not used to define the topology; it is a geometric realization suggested by the Fourier coordinates.

The topology is verified combinatorially. Let $\mathcal{F}$ consist of the nine square 4-cycles of the 3×3 periodic grid. The dart condition is satisfied, and hence

$$F = 9.$$

Therefore

$$\chi = V - E - F + 2F = V - E + F = 9 - 18 + 9 = 0.$$

Equivalently,

$$\chi = V - E + F = 0.$$

By the genus formula

$$\chi = 2 - 2g,$$

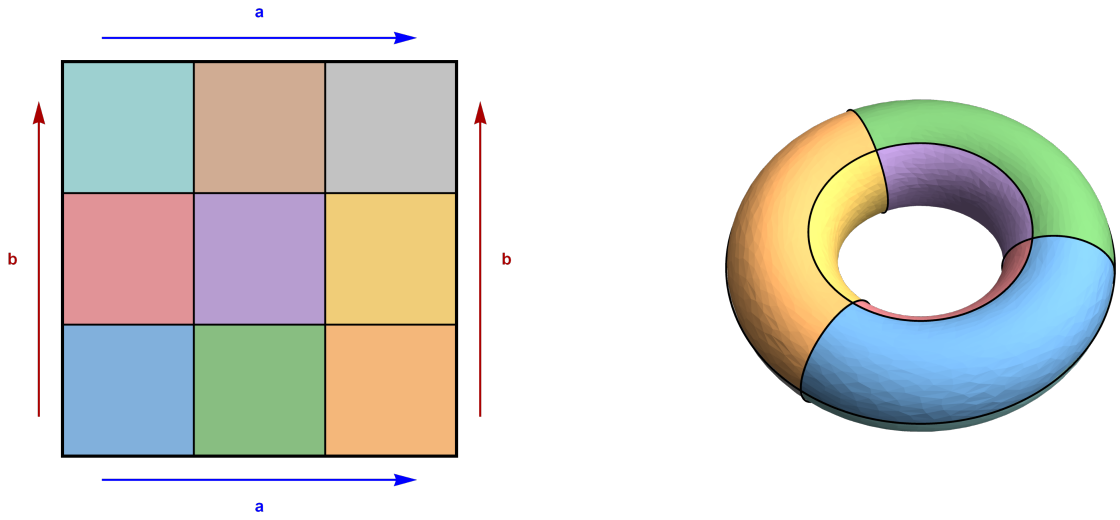


Figure 1: Fourier torus realization of $\mathrm{Cay}(\mathbb{Z}_3 \times \mathbb{Z}_3, S)$ with $S = \{(\pm 1, 0), (0, \pm 1)\}$. The left panel shows the 3×3 periodic fundamental domain, where the blue and red arrows indicate the two independent cyclic directions of the group. The nine colored squares correspond to the nine elements of $\mathbb{Z}_3 \times \mathbb{Z}_3$. The right panel shows the corresponding smoothed toroidal realization in $\mathbb{R}^3$. Spectrally, the two cyclic directions give the direct-sum coordinate structure $\mathbb{R}^2 \oplus \mathbb{R}^2$, which is geometrically interpreted as two angular phase coordinates on $S^1 \times S^1$. The face system consists of nine square 4-cycles, giving $\chi = 9 - 18 + 9 = 0$, hence a torus.

we obtain

$$g = 1.$$

Thus the verified surface is a torus.

This example illustrates the cleanest toroidal signal in the thesis: a $2+2$ Fourier coordinate structure suggests a torus, and the nine-square face system verifies it through the dart condition and the Euler characteristic.

4.3.2 The Dihedral Group D_4 and Topological Switching under a Static Eigenspace

We now consider the non-abelian example

$$D_4 = \langle r, s \mid r^4 = e,\ s^2 = e,\ srs = r^{-1} \rangle$$

with symmetric generating set

$$S = \{r, r^{-1}, s\}.$$

The Cayley graph $\mathrm{Cay}(D_4, S)$ has

$$V = |D_4| = 8, \qquad E = \frac{|D_4||S|}{2} = \frac{8 \cdot 3}{2} = 12.$$

Its unweighted graph Laplacian is

$$L = |S|I - A = 3I - A.$$

The irreducible representations of D_4 consist of four one-dimensional representations and one two-dimensional representation. Let the one-dimensional representations be denoted by

$$\rho_1, \rho_2, \rho_3, \rho_4,$$

and let the two-dimensional representation be denoted by ρ_5. A standard choice is

$$\rho_1(r) = 1, \quad \rho_1(s) = 1,$$

$$\rho_2(r) = 1, \quad \rho_2(s) = -1,$$

$$\rho_3(r) = -1, \quad \rho_3(s) = 1,$$

$$\rho_4(r) = -1, \quad \rho_4(s) = -1,$$

and

$$\rho_5(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \rho_5(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For each representation ρ, the adjacency block is

$$A_\rho = \sum_{t \in S} \rho(t) = \rho(r) + \rho(r^{-1}) + \rho(s).$$

Computing these blocks gives the Laplacian spectrum

$$\mathrm{Spec}(L) = \{0^{(1)},\ 2^{(3)},\ 4^{(3)},\ 6^{(1)}\}.$$

The lowest non-zero eigenspace has eigenvalue

$$\lambda = 2$$

and dimension 3. Its representation-theoretic structure is

$$\mathcal{V}_{\lambda=2} = V_{\rho_2} \oplus V_{\rho_5}.$$

This is a direct sum of a one-dimensional component and a two-dimensional component.

Geometrically, the V_{ρ_5} component gives the fourfold rotational structure in a plane, while the V_{ρ_2} component separates the vertices into two parallel layers. Together they produce a fixed three-dimensional spectral drawing of the Cayley skeleton $\mathrm{Cay}(D_4, S)$.

The crucial point is that this spectral drawing is fixed once (D_4, S) is fixed. It does not change when the face system changes. However, the topology of the resulting surface depends entirely on the choice of $\mathcal{F}$.

The 4-cycle face system. Choose $\mathcal{F}$ to consist of six square faces. The dart condition is satisfied, and therefore

$$F = 6.$$

Since

$$V = 8, \qquad E = 12,$$

we obtain

$$\chi = V - E + F = 8 - 12 + 6 = 2.$$

Hence

$$g = 1 - \frac{\chi}{2} = 1 - \frac{2}{2} = 0.$$

The resulting closed orientable surface is the sphere S^2. Geometrically, the same Cayley skeleton is interpreted as the one-dimensional skeleton of a cube.

The 6-cycle face system. Choose instead a face system $\mathcal{F}$ consisting of four hexagonal faces. Again, the dart condition is satisfied on the same dart set $D_4 \times S$. Now

$$F = 4,$$

so

$$\chi = V - E + F = 8 - 12 + 4 = 0.$$

Hence

$$g = 1.$$

The resulting closed orientable surface is the torus T^2.

This example is the central topological switching phenomenon of the thesis. The graph $\mathrm{Cay}(D_4, S)$, the adjacency matrix A, the Laplacian

$$L = |S|I - A,$$

and the spectral embedding space

$$\mathcal{V}_{\lambda=2} = V_{\rho_2} \oplus V_{\rho_5}$$

are all fixed. Nevertheless, two different valid face systems produce two different genera. Therefore the topology is not determined by the vertex spectrum. It is determined by the combinatorial choice of $\mathcal{F}$.

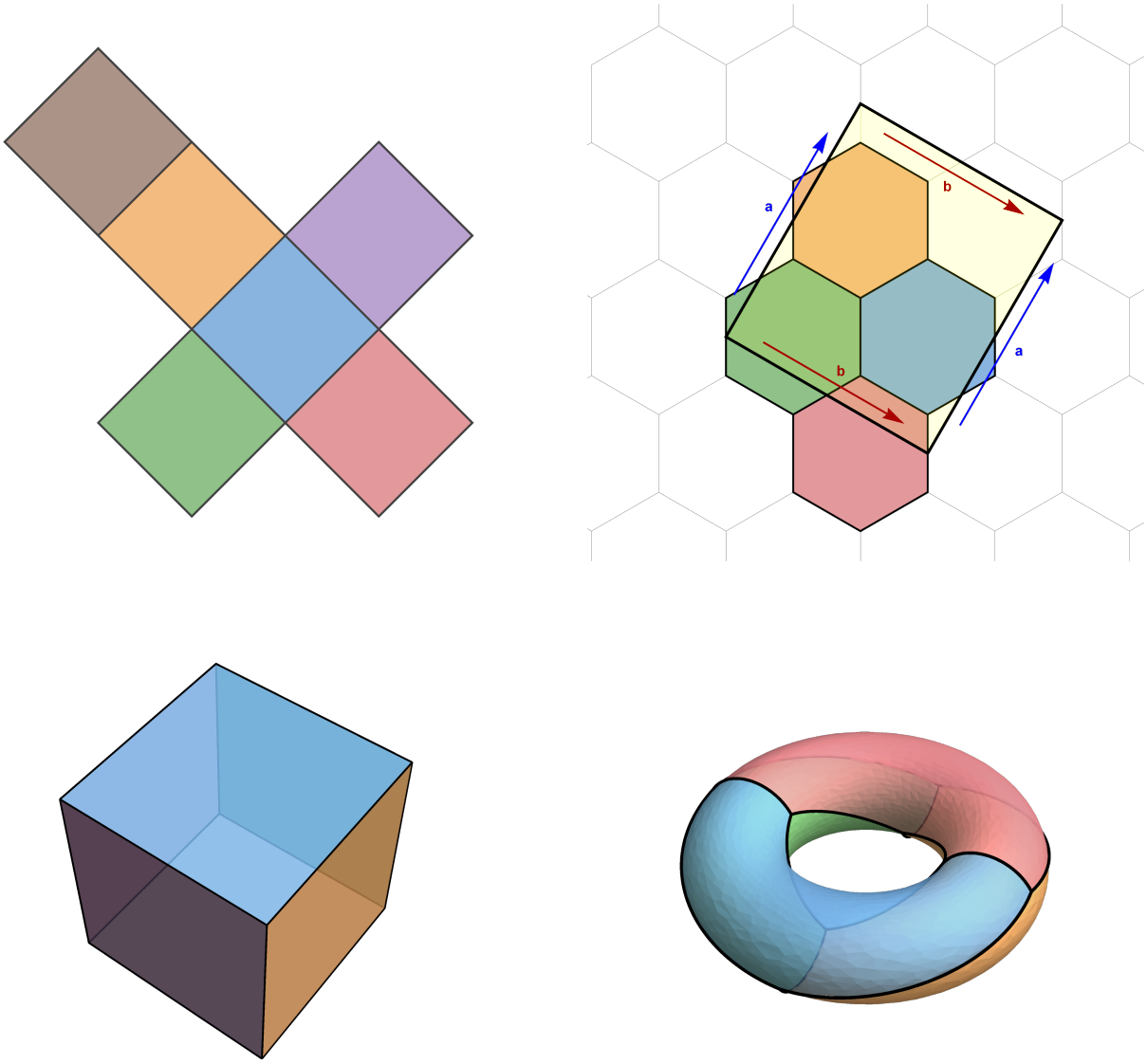


Figure 2: Topological switching on the fixed Cayley skeleton $\mathrm{Cay}(D_4, \{r, r^{-1}, s\})$. The left column shows the 4-cycle face system: six square faces are assembled into a cubical realization of the sphere, with $V = 8$, $E = 12$, $F = 6$, and $\chi = 8 - 12 + 6 = 2$. The right column shows the 6-cycle face system: four hexagonal faces are assembled into a toroidal realization, with $V = 8$, $E = 12$, $F = 4$, and $\chi = 8 - 12 + 4 = 0$. In both cases, the Cayley skeleton and the unweighted Laplacian $L = |S|I - A$ are fixed. The topological difference is entirely determined by the choice of face system $\mathcal{F}$.

5 Dual Graphs, Face Inflation, and Topologically Sensitive Invariants

The previous section showed that the vertex Laplacian

$$L = |S|I - A$$

is determined entirely by the Cayley skeleton $\mathrm{Cay}(G, S)$. It therefore does not change when the face system $\mathcal{F}$ changes. This is the spectral blindness of vertex operators: they detect the symmetry of the one-dimensional skeleton, but not the two-dimensional combinatorial choice of faces.

To record information depending on $\mathcal{F}$, we now shift from vertex adjacency to face adjacency and then refine this idea further by introducing the face-inflated graph. The goal is to construct graph objects whose spectra are sensitive to the chosen surface system while remaining within the same discrete linear-algebraic framework.

5.1 The Spectral Blindness of Vertex Operators and Classical Duality

Fix a finite group G, a symmetric generating set S, and the Cayley skeleton

$$\Gamma = \mathrm{Cay}(G, S).$$

Let A be the unweighted adjacency matrix of Γ, and let

$$L = |S|I - A$$

be its unweighted graph Laplacian. Since A is defined only from the edges of $\mathrm{Cay}(G, S)$, the operator L depends only on the pair (G, S). It does not depend on any face system $\mathcal{F}$.

Thus, if the same Cayley graph admits two different valid face systems $\mathcal{F}_1$ and $\mathcal{F}_2$, then the vertex Laplacian remains the same:

$$L_{\mathcal{F}_1} = L_{\mathcal{F}_2} = L_{\mathrm{Cay}(G,S)}.$$

Consequently,

$$\mathrm{Spec}(L_{\mathcal{F}_1}) = \mathrm{Spec}(L_{\mathcal{F}_2}) = \mathrm{Spec}(L_{\mathrm{Cay}(G,S)}).$$

This equality is not a topological statement; it simply reflects the fact that the vertex operator has no access to the chosen faces.

The classical way to introduce face-level information is to form the dual graph.

Definition 5.1 (Classical dual graph)**.** Let $(G, S, \mathcal{F})$ be a valid Cayley surface system. The classical dual graph, denoted by

$$\Gamma^* = (V^*, E^*),$$

is defined as follows:

(i) for each face $F \in \mathcal{F}$, introduce one dual vertex $v_F^* \in V^*$;

(ii) if two faces $F, F' \in \mathcal{F}$ share an edge of the original Cayley skeleton, then place a dual edge between v_F^* and $v_{F'}^*$.

Equivalently, suppose a dart

$$(g, s)$$

belongs to the face F, while its reversal

$$\overline{(g, s)} = (gs, s^{-1})$$

belongs to the face F'. Then the original edge $\{g, gs\}$ contributes an adjacency between the dual vertices v_F^* and $v_{F'}^*$. If the same pair of faces share more than one original edge, the dual graph may be treated as a multigraph or, after aggregation, as a weighted graph.

Let A^* be the adjacency matrix of Γ^*, and let

$$L^* = D^* - A^*$$

be its unweighted dual Laplacian when Γ^* is treated as a simple graph, or the corresponding weighted Laplacian when multiple shared edges are aggregated. Unlike the vertex Laplacian L, the dual operators A^* and L^* depend explicitly on the face system $\mathcal{F}$.

This already distinguishes the two D_4 surface systems. For

$$G = D_4, \qquad S = \{r, r^{-1}, s\},$$

the Cayley skeleton has

$$V = 8, \qquad E = 12.$$

The square-face system has

$$F = 6, \qquad \chi = 8 - 12 + 6 = 2,$$

and its dual graph has six dual vertices. Geometrically, this is the dual structure associated with the cubical sphere system. The hexagon-face system has

$$F = 4, \qquad \chi = 8 - 12 + 4 = 0,$$

and its dual graph has four dual vertices. Thus the dual graph changes when the face system changes, even though the original vertex spectrum does not.

The classical dual graph therefore partially resolves the spectral blindness of L. However, it does so by compressing every face into a single dual vertex, and this compression loses important information. This motivates the refined constructions in the next subsection.

5.2 Information Loss and the Construction of Face-Inflated Graphs

The classical dual graph Γ^* records which faces are adjacent, but it does not faithfully record the full surface system. Two types of information are lost.

First, Γ^* forgets the original Cayley vertices. Its vertices are faces, not group elements. Therefore the explicit vertex identity coming from the group G is no longer visible.

Second, Γ^* forgets the cyclic dart order inside each face. It records that two faces are adjacent, but not the order in which darts appear around a face. Since the dart order is precisely what encodes the local orientation of each face cycle, this information is essential for reconstructing the chosen surface system.

To reduce this loss, we introduce two augmented graph structures. The first is the face-center augmented graph. The second, more refined construction is the face-inflated graph

$$\Gamma_{\mathrm{inf}}.$$

5.2.1 The Face-Center Augmented Graph

The face-center augmented graph keeps the original Cayley skeleton and adds one new vertex for each face.

Definition 5.2 (Face-center augmented graph)**.** Let $(G, S, \mathcal{F})$ be a valid Cayley surface system. The *face-center augmented graph*, denoted by

$$\Gamma_{\mathrm{aug}},$$

is constructed as follows:

(i) retain all original vertices v_g of $\mathrm{Cay}(G, S)$;

(ii) retain all original Cayley edges $\{v_g, v_{gs}\}$;

(iii) for each face $F \in \mathcal{F}$, add one new vertex c_F;

(iv) if the original vertex v_g appears in the cyclic vertex list of the face F, add an incidence edge between v_g and c_F.

This construction records which original vertices occur in which faces. It therefore preserves more information than the classical dual graph. It also keeps the original Cayley skeleton visible, so the left action of G remains explicit when $\mathcal{F}$ is G-invariant.

However, Γ_{aug} still compresses each face into one center vertex. It records vertex-face incidence, but it does not record the cyclic order of darts inside the face. Therefore it is still not the most faithful graph-theoretic encoding of the surface system.

5.2.2 The Face-Inflated Graph

The face-inflated graph is designed to preserve the original vertex identities, the chosen faces, and the cyclic dart order simultaneously. Instead of collapsing a face into one vertex, it replaces each face by a small ordered polygon.

Definition 5.3 (Face-inflated graph)**.** Let $(G, S, \mathcal{F})$ be a valid Cayley surface system satisfying the dart condition. For each face

$$F = (v_1, v_2, \ldots, v_k) \in \mathcal{F},$$

where the order is the cyclic order determined by the dart sequence of F, introduce face-local vertices

$$u_{F,1}, u_{F,2}, \dots, u_{F,k}.$$

The face-inflated graph

$$\Gamma_{\mathrm{inf}}$$

is constructed by the following rules.

(i) **Original Cayley skeleton.** Retain the original vertices v_g and the original edges of $\mathrm{Cay}(G, S)$.

(ii) **Face-local copies.** For every occurrence of a vertex v_i in a face $F = (v_1, \dots, v_k)$, add a local copy $u_{F,i}$.

(iii) **Intra-face cyclic edges.** Add edges

$$u_{F,i} \sim u_{F,i+1}, \qquad i = 1, \dots, k,$$

with indices taken modulo k. These edges record the cyclic dart order of the face.

(iv) **Radial incidence edges.** Add edges

$$v_i \sim u_{F,i}, \qquad i = 1, \dots, k.$$

These edges attach each face-local copy back to its corresponding original Cayley vertex.

The face-inflated graph therefore contains three layers of information:

$$\mathrm{Cay}(G, S), \qquad \mathcal{F}, \qquad \text{the cyclic dart order in each face.}$$

This makes Γ_{inf} more informative than the classical dual graph Γ^* and the face-center augmented graph Γ_{aug}.

If $\mathcal{F}$ is invariant under the left action of G, then Γ_{inf} also inherits a natural G-action. The original vertices transform by

$$h \cdot v_g = v_{hg},$$

and the local copy attached to the face F transforms to the corresponding local copy attached to the translated face $h \cdot F$. Thus the inflated graph remains compatible with the algebraic symmetry of the Cayley skeleton.

For spectral analysis, one may form the unweighted adjacency matrix A_{inf} of Γ_{inf} and the corresponding unweighted Laplacian

$$L_{\mathrm{inf}} = D_{\mathrm{inf}} - A_{\mathrm{inf}}.$$

This operator is no longer blind to $\mathcal{F}$, because the graph Γ_{inf} itself was built from the chosen face system. Thus L_{inf} provides a matrix-level object that simultaneously records the skeleton and the face structure.

The role of Γ_{inf} can be summarized as follows:

the Cayley graph gives the skeleton,

the face system verifies the surface,

and the face-inflated graph records the verified structure.

Face-Inflated Graph Placeholder

Classical dual graph Γ^*: one vertex for each face.
Face-inflated graph Γ_{inf}: each face is replaced by an ordered polygon of local vertex copies $(u_{F,1}, \ldots, u_{F,k})$, with radial incidence edges back to the original Cayley vertices.

Figure 3: Comparison between the classical dual graph Γ^* and the face-inflated graph Γ_{inf}. In the classical dual graph, each face of the surface system $\mathcal{F}$ is collapsed to a single vertex, so the original Cayley vertices and the cyclic dart order are not retained. In the face-inflated graph, each face $F = (v_1, \ldots, v_k)$ is replaced by a small ordered polygon $(u_{F,1}, \ldots, u_{F,k})$. The intra-face polygon edges record the cyclic dart order, while the radial incidence edges attach each local copy $u_{F,i}$ back to the corresponding original Cayley vertex v_i. This structure preserves face adjacency, vertex identity, and cyclic order simultaneously.

5.3 Case Study on Q_8: Non-Abelian Toroidal Surface System

We now examine the quaternion group

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$$

with symmetric generating set

$$S = \{i, -i, j, -j\}.$$

The Cayley graph

$$\text{Cay}(Q_8, S)$$

has

$$V = |Q_8| = 8, \qquad |S| = 4,$$

and hence

$$E = \frac{|Q_8||S|}{2} = \frac{8 \cdot 4}{2} = 16.$$

The unweighted graph Laplacian is

$$L = |S|I - A = 4I - A.$$

The group Q_8 has four one-dimensional irreducible representations and one two-dimensional irreducible representation. For the generating set $S = \{i, -i, j, -j\}$, the adjacency spectrum is obtained from the matrices

$$A_\rho = \sum_{s \in S} \rho(s).$$

The trivial representation contributes adjacency eigenvalue 4, one one-dimensional representation contributes adjacency eigenvalue -4, and the remaining components contribute adjacency eigenvalue 0. Therefore the Laplacian spectrum is

$$\text{Spec}(L) = \{0^{(1)}, 4^{(6)}, 8^{(1)}\}.$$

The lowest non-zero eigenvalue is

$$\lambda_L = 4.$$

Although the full eigenspace for $\lambda_L = 4$ has dimension 6, a three-dimensional invariant coordinate subspace can be selected for visualization. In the spectral drawing used here, we take a direct sum of one real one-dimensional component and one real two-dimensional component:

$$W_{\text{draw}} = V_{\rho_a} \oplus V_{\rho_5}^{\mathbb{R}},$$

where $V_{\rho_5}^{\mathbb{R}}$ denotes the real coordinate space coming from the two-dimensional representation. This gives a three-dimensional symmetry-preserving drawing

$$\Phi : Q_8 \to \mathbb{R}^3.$$

One normalized coordinate realization is:

$$\Phi(1) = \left(\frac{1}{2}, 0, \frac{1}{2\sqrt{2}}\right), \qquad \Phi(-1) = \left(-\frac{1}{2}, 0, \frac{1}{2\sqrt{2}}\right),$$

$$\Phi(i) = \left(0, \frac{1}{2}, \frac{1}{2\sqrt{2}}\right), \qquad \Phi(-i) = \left(0, -\frac{1}{2}, \frac{1}{2\sqrt{2}}\right),$$

$$\Phi(j) = \left(0, \frac{1}{2}, -\frac{1}{2\sqrt{2}}\right), \qquad \Phi(-j) = \left(0, -\frac{1}{2}, -\frac{1}{2\sqrt{2}}\right),$$

$$\Phi(k) = \left(\frac{1}{2}, 0, -\frac{1}{2\sqrt{2}}\right), \qquad \Phi(-k) = \left(-\frac{1}{2}, 0, -\frac{1}{2\sqrt{2}}\right).$$

Equivalently, using decimal approximations,

$$\frac{1}{2\sqrt{2}} \approx 0.3536.$$

We now choose a valid face system $\mathcal{F}$ on this fixed Cayley skeleton. Since

$$E = 16,$$

the dart set has size

$$|\mathcal{D}| = 2E = 32.$$

Therefore a toroidal face system with

$$F = 8$$

must have total face length 32. In the quadrilateral case this means that $\mathcal{F}$ consists of eight 4-cycles.

One such G-invariant quadrilateral structure is generated by the relator cycles

$$i^4 = e, \qquad (-i)^4 = e, \qquad j^4 = e, \qquad (-j)^4 = e,$$

together with their left translates. These cycles give a collection of eight oriented square faces. Their dart sequences exactly partition

$$\mathcal{D} = Q_8 \times S.$$

Hence the dart condition is satisfied.

The Euler characteristic is

$$\chi = V - E + F = 8 - 16 + 8 = 0.$$

By the classification of closed orientable surfaces,

$$\chi = 2 - 2g.$$

Thus

$$0 = 2 - 2g, \qquad g = 1.$$

The resulting surface is therefore a torus.

The significance of this example is twofold. First, the vertex spectrum of $L = 4I - A$ is fixed by the pair (Q_8, S) and does not determine the face system. Second, once the quadrilateral face system $\mathcal{F}$ is chosen and verified by the dart condition, the face-inflated graph Γ_{inf} records the resulting non-abelian toroidal surface structure in a graph-theoretic form suitable for further spectral analysis.

Thus the Q_8 example reinforces the main thesis:

spectral drawing suggests, face systems verify, and

face-inflated graphs record.

6 Conclusions and Future Directions

This thesis develops a unified algebraic and discrete-geometric framework for constructing closed orientable surfaces from Cayley graphs. Starting from a finite group G and a symmetric generating set S, the Cayley graph

$$\mathrm{Cay}(G, S)$$

provides a highly symmetric one-dimensional skeleton. However, this skeleton alone does not determine a surface. The decisive additional datum is the face system $\mathcal{F}$.

The main conclusion of the thesis is that the topology of the resulting surface is determined by the triple

$$(G, S, \mathcal{F}),$$

not by the pair (G, S) alone. The graph Laplacian

$$L = |S|I - A$$

is a rigid spectral invariant of the Cayley skeleton, but it is blind to the choice of $\mathcal{F}$. Therefore spectral drawing can suggest natural geometric realizations, but the genus of the surface is verified only after the dart condition is checked and the Euler characteristic

$$\chi = V - E + F$$

is computed.

The conceptual structure of the thesis can be summarized by the guiding principle

spectral drawing suggests; face systems verify; face-inflated graphs record.

Spectral drawing extracts symmetry-preserving coordinates from eigenspaces of L. Face systems verify whether the suggested cycles actually assemble into a closed orientable surface. Face-inflated graphs then record the verified surface system while preserving vertex identity, face incidence, and cyclic dart order.

6.1 Summary of Principal Contributions

The main contributions of this thesis are organized into three parts.

(1) **Separation of geometry and topology.**

The thesis shows that geometric embedding and topological classification are logically distinct. The spectral embedding of $\mathrm{Cay}(G, S)$ is determined by the unweighted Laplacian

$$L = |S|I - A.$$

Hence it depends only on the group-generator pair (G, S). By contrast, the genus depends on the face count

$$F = |\mathcal{F}|.$$

Thus the surface type depends on the full triple $(G, S, \mathcal{F})$.

The D_4 example demonstrates this separation explicitly. The same Cayley skeleton with

$$V = 8, \qquad E = 12$$

admits a square-face system with

$$F = 6, \qquad \chi = 8 - 12 + 6 = 2,$$

producing a sphere, and a hexagon-face system with

$$F = 4, \qquad \chi = 8 - 12 + 4 = 0,$$

producing a torus. The graph and its vertex spectrum remain fixed; only the face system changes.

(2) **Dart formalism for closed orientable surface verification.**

The thesis replaces informal geometric intuition with an exact combinatorial test. A dart is a directed edge

$$(g, s) \in G \times S,$$

with reversal

$$\overline{(g, s)} = (gs, s^{-1}).$$

A face system $\mathcal{F}$ is valid precisely when every dart belongs to exactly one face. This dart condition simultaneously verifies the two necessary surface properties: every edge is paired by opposite dart directions, and the cyclic orientations of adjacent faces are compatible.

This gives a purely discrete verification principle:

$$\mathcal{F} \text{ is valid} \quad \Longleftrightarrow \quad \mathcal{F} \text{ exactly partitions } G \times S.$$

Once this condition is verified, the topological type follows from

$$\chi = V - E + F, \qquad \chi = 2 - 2g.$$

(3) **Face-inflated graphs as topology-sensitive spectral data.**

The classical dual graph Γ^* records face adjacency, but it compresses each face into a single dual vertex. As a result, it loses the original Cayley vertex identities and the cyclic dart order inside each face.

To retain this information, the thesis introduces the face-inflated graph

$$\Gamma_{\text{inf}}.$$

In Γ_{inf}, each face is replaced by a small ordered polygon whose vertices are face-local copies of the original Cayley vertices. The polygon edges record the cyclic dart order, and radial incidence edges attach each local copy back to the original Cayley skeleton.

Therefore Γ_{inf} records three layers of structure simultaneously:

$$\text{Cay}(G, S), \qquad \mathcal{F}, \qquad \text{cyclic dart order.}$$

This makes it a refined graph-theoretic object for studying a chosen surface system through spectral methods.

Together, these contributions establish a consistent pipeline from finite group data to surface topology. The Cayley graph supplies the symmetric skeleton, the face system supplies the topological filling, and the face-inflated graph supplies a topology-sensitive spectral object.

6.2 Open Problems and Future Work

The framework developed in this thesis suggests several directions for future research.

6.2.1 Hyperbolic Embeddings and Higher-Genus Surfaces

The examples treated in this thesis mainly involve spherical and toroidal surface types. The spherical case corresponds to

$$\chi = 2, \qquad g = 0,$$

while the toroidal case corresponds to

$$\chi = 0, \qquad g = 1.$$

For higher-genus surfaces,

$$g \geq 2, \qquad \chi = 2 - 2g < 0,$$

the natural geometric setting is hyperbolic rather than Euclidean.

A major future direction is to extend the spectral drawing pipeline to hyperbolic models. In this setting, low-frequency eigenspaces of

$$L = |S|I - A$$

may still provide symmetry-preserving coordinates, but the final geometric realization should respect the negative-curvature nature of higher-genus surfaces. Possible directions include embeddings into the Poincaré disk model or other discrete hyperbolic coordinate systems.

The central question is:

Can the representation-theoretic spectrum of $\mathrm{Cay}(G, S)$ *guide hyperbolic embeddings of valid higher-genus*

6.2.2 Spectral Uncertainty Bounds and Expander Families

Another direction concerns the relationship between face systems and spectral gaps. For a graph Laplacian, the first non-zero eigenvalue controls connectivity and expansion behavior. In the present framework, there are at least two relevant spectral objects:

$$L = |S|I - A$$

for the original Cayley skeleton, and

$$L_{\mathrm{inf}} = D_{\mathrm{inf}} - A_{\mathrm{inf}}$$

for the face-inflated graph.

The vertex Laplacian L depends only on (G, S), while L_{inf} depends on the full surface system $(G, S, \mathcal{F})$. This raises a natural question:

How does the choice of $\mathcal{F}$ *affect the spectral gap of* Γ_{inf}?

One possible research direction is to investigate whether certain face systems maximize or minimize the spectral gap of the inflated graph. If such extremal behavior can be related to representation-theoretic constraints on G, then the construction may connect Cayley surface systems with the theory of expander graphs.

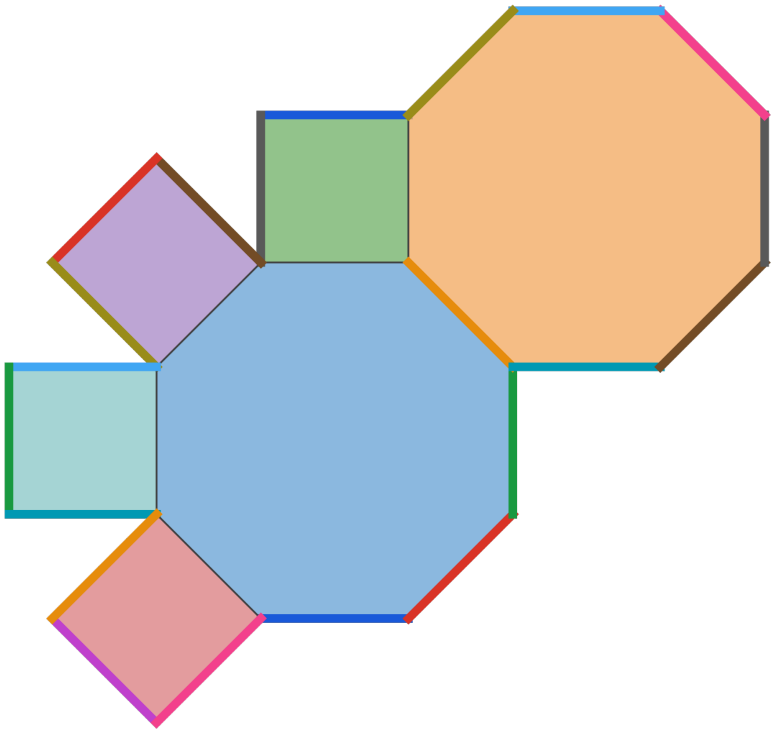


Figure 4: A higher-genus face-domain pattern associated with a Cayley surface system of the quaternion group Q_8. The mixture of polygonal face types suggests a more complicated surface system than the toroidal quadrilateral case treated in Section 5.3. This example illustrates why higher-genus Cayley surfaces require additional geometric tools, such as hyperbolic embedding methods, beyond the spherical and toroidal realizations developed in the main examples.

6.2.3 Geometry-Guided Automated Face Search

A practical limitation of the current framework is the search for valid face systems. A candidate face system is produced by selecting relator cycles, translating them by the left G-action, and checking whether the resulting dart sequences exactly partition

$$G \times S.$$

For large groups, this search can become computationally expensive.

A future algorithmic direction is to use spectral drawing as a geometric guide for face search. Instead of enumerating all relator cycles purely algebraically, one may first embed the Cayley skeleton using low-frequency eigenspaces of

$$L = |S|I - A.$$

The resulting geometry may suggest which cycles are likely to bound natural faces. These cycles can then be tested by the dart condition.

This leads to a geometry-guided pipeline:

$$\text{spectral drawing} \longrightarrow \text{candidate cycles} \longrightarrow \text{dart verification} \longrightarrow \text{surface classification}.$$

Such a method would make the construction more scalable and could allow systematic exploration of larger non-abelian groups.

6.2.4 Classification of Face Systems on a Fixed Cayley Skeleton

The D_4 example shows that a single Cayley skeleton may admit multiple valid face systems with different genera. This suggests a broader classification problem.

For a fixed pair (G, S), define

$$\mathfrak{F}(G, S)$$

to be the collection of all G-invariant face systems satisfying the dart condition. Each element $\mathcal{F} \in \mathfrak{F}(G, S)$ determines a surface genus by

$$g(\mathcal{F}) = 1 - \frac{|G| - \frac{|G||S|}{2} + |\mathcal{F}|}{2}.$$

The classification problem asks:

What genera can occur among valid face systems on a fixed $\mathrm{Cay}(G, S)$?

This question isolates the combinatorial freedom of the face system. It also clarifies the central distinction of the thesis: the Cayley skeleton fixes the graph, but the face system determines the topology.

6.2.5 Dual and Inflated Spectra as Surface Invariants

Finally, the face-inflated graph suggests a new family of topology-sensitive spectral invariants. The classical dual graph Γ^* records face adjacency, while Γ_{inf} records face adjacency together with vertex identity and cyclic dart order. Comparing their spectra may reveal how much topological information is retained at each level of encoding.

A possible future program is to study the relationship among

$$\mathrm{Spec}(L), \qquad \mathrm{Spec}(L^*), \qquad \mathrm{Spec}(L_{\mathrm{inf}}),$$

where L is the vertex Laplacian of $\mathrm{Cay}(G, S)$, L^* is the dual Laplacian of Γ^*, and L_{inf} is the Laplacian of the face-inflated graph. Since L depends only on (G, S) while L^* and L_{inf} depend on $\mathcal{F}$, these spectra separate the skeleton-level and surface-level information.

The guiding question is:

Which spectral features of Γ_{inf} *are genuine invariants of the chosen Cayley surface system?*

Answering this question would strengthen the role of face-inflated graphs as a bridge between algebraic symmetry, discrete geometry, and surface topology.